

Program 5:

Demonstrate CI Integration by running Jenkins in a Docker container

21/11/2025

Project Structure:

```
prog5
|- Dockerfile
|- docker-compose.yaml
```

Steps for Execution:

Step 1: Create a directory with the prog3 name in order to execute the program

```
> mkdir prog5
```

```
> cd prog5/
```

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~$ cd devops_lab/
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab$ mkdir prog5
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab$ cd prog5/
```

Step 2: Create a two files,

1. Dockerfile 2. docker-compose.yaml

The content which has to be included in those files is shown in below images

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog5$ nano Dockerfile
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog5$ cat Dockerfile
FROM jenkins/jenkins:lts

# Switch to root to install packages
USER root

# Install git (Optional: install docker client too)
RUN apt-get update && \
    apt-get install -y git docker.io && \
    rm -rf /var/lib/apt/lists/*

# Switch back to Jenkins user
USER jenkins

1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog5$ nano docker-compose.yml
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog5$ cat docker-compose.yml
version: "3.8"
```

```
services:
  jenkins:
    build: .
    container_name: program5-jenkins
    ports:
      - "8080:8080"
      - "50000:50000"
    volumes:
      - jenkins_home:/var/jenkins_home
      - /var/run/docker.sock:/var/run/docker.sock

volumes:
  jenkins_home:
```

Step 3: Build the container by using the command,

> docker compose build

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog5$ docker compose build
WARN[0000] /home/chinnu/devops_lab/prog5/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Building 90.4s (9/9) FINISHED
=> [internal] load local bake definitions                                0.0s
=> => reading from stdin 511B                                           0.0s
=> [internal] load build definition from Dockerfile                     0.0s
=> => transferring dockerfile: 307B                                     0.0s
=> [internal] load metadata for docker.io/jenkins/jenkins:ls         9.2s
=> [auth] jenkins/jenkins:pull token for registry-1.docker.io        0.0s
=> [internal] load .dockerignore                                       0.0s
=> => transferring context: 2B                                          0.0s
=> [1/2] FROM docker.io/jenkins/jenkins:ls@sha256:1e407df1a9bcb122cc4d 24.6s
=> => resolve docker.io/jenkins/jenkins:ls@sha256:1e407df1a9bcb122cc4d7 0.0s
=> => sha256:d2519f76d4c6527e03084f2742da3ae15b0c5097d464be4 392B / 392B 1.0s
=> => sha256:986be6969b0a04975a2e559b30a96d766bf360d0b34 1.29kB / 1.29kB 1.4s
=> => sha256:3fb70446889ebcc58e91f43f90aafa314798941c318 2.23kB / 2.23kB 2.2s
=> => sha256:d7dd2e63023897146ec4b913a460382968fbd1c 72.05MB / 72.05MB 23.6s
=> => sha256:a9230f7c9c31780c6b1dc4f2aa7e1073df2fdf88e9c 6.29MB / 6.29MB 2.4s
=> => sha256:501deb283b09e30fd3f1f4e27a635bcc1739485e8ba8050 189B / 189B 1.7s
=> => sha256:687044b042c5d6d57e2ce207c000ee159f2382cb 95.32MB / 95.32MB 18.4s
=> => sha256:e0e73f34dee93322ae4db1d44250aab1f61011a2b7ec954 183B / 183B 0.5s
=> => sha256:68cdef48009293f6a12f5c50b91fcfe4194bd629333 1.23kB / 1.23kB 0.5s
=> => sha256:2f1ecb09fa362ebe652038ecbfcc64a84b042991e9d 5.24MB / 5.24MB 2.0s
=> => sha256:5687d4df0728e358596e7d718163b2e812a00cf5 65.08MB / 65.08MB 9.3s
=> => extracting sha256:5687d4df0728e358596e7d718163b2e812a00cf5dd26244 1.6s
```

Step 4: Now up the image so that it can run by using the command,

> docker compose up -d

Step 5: Now check the docker images and containers which is created by using the command,

> docker images

> docker ps

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog5$ docker compose up -d
WARN[0000] /home/chinnu/devops_lab/prog5/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Running 1/1
✔ Container program5-jenkins Started                                0.4s
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog5$ docker images
IMAGE                                ID                                DISK USAGE  CONTENT SIZE  EXTRA
prog5-jenkins:latest                013965e6e471                    1.39GB      438MB        In Use
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog5$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
ca5ac20a3895   prog5-jenkins                       "/usr/bin/tini -- /u..." 13 seconds ago Up 12 seconds 0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp
```

Step 6: Open any browser and search for,

> http://localhost:8080/

